

Corrected Max-Rank for Finite-Sample Conformal Prediction

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Abstract

Max-Rank is a dependence-aware family-wise error rate correction method used to construct hyperrectangular prediction regions in conformal multi-target regression. It ranks dimension-wise internal nonconformity scores and uses the largest dimension-wise rank as a joint nonconformity score, often achieving smaller regions than Bonferroni correction. In this paper, we revisit the finite-sample validity of Max-Rank and distinguish between the Max-Rank rule in the rank space and its implementation in the dimension-wise score space. The rank-space rule is valid. Its original score-space implementation, however, is incorrect and loses the finite-sample coverage guarantee. The implementation translates a rank threshold q into the q th dimension-wise calibration score. We show that this threshold is one order statistic too small and that the correct score threshold is the $(q + 1)$ st dimension-wise calibration score, with the usual $+\infty$ convention when no large enough order statistic exists. We propose Corrected Max-Rank, a straightforward modification that uses the correct rank-to-score implementation. We prove that the corrected implementation satisfies the desired finite-sample coverage guarantee under exchangeability and dimension-separable internal scores. Experiments on synthetic multidimensional internal score distributions confirm that the original implementation can exceed the targeted error rate, especially for small calibration sets and a larger number of output dimensions. Corrected Max-Rank restores validity at the cost of slightly wider prediction regions.

Keywords: inductive conformal prediction, multi-target regression, hyperrectangular prediction regions, Max-Rank, joint coverage

1. Introduction

Conformal prediction provides prediction regions with finite-sample validity guarantees under exchangeability (Vovk et al., 2022). In ordinary regression, the output is usually a single number and the conformal predictor returns one interval. In multi-target regression, however, the output is a vector with several dimensions. In this case, the prediction region should contain the true label in all output dimensions simultaneously with probability at least $1 - \epsilon$ (Timans et al., 2025; Vovk, 2013).

An easily interpretable prediction region for multi-target regression is an axis-aligned hyperrectangle. The conformal predictor returns one interval per output dimension, and the Cartesian product of these intervals forms the prediction region. The difficulty is that

marginally valid intervals for each dimension do not automatically give a jointly valid hyperrectangle. Some correction is needed to control the probability that the true label is not contained in the prediction region in at least one dimension.

A standard solution is the Bonferroni correction (Vovk, 2013). Bonferroni is simple and valid, but it can be conservative because it does not exploit dependence between the output dimensions. If the dimension-wise nonconformity scores are correlated, the Bonferroni correction produces unnecessarily wide intervals (Schlembach et al., 2025).

Max-Rank, introduced by Timans et al. (2025), is an elegant dependence-aware alternative. First, for every calibration instance and every output dimension, compute an internal nonconformity score in the *score space*. An example of an internal nonconformity score is the dimension-wise absolute residual. Second, within each dimension, rank the internal nonconformity scores in ascending order. Third, for each calibration instance, take the largest associated rank over all dimensions. This maximum rank is the **Max-Rank** score of that calibration instance in the *rank space*. A calibration instance has a large **Max-Rank** score if it has an extreme internal nonconformity score in at least one output dimension.

For a new test instance, the *rank-space Max-Rank rule* defines which candidate labels are included in the prediction set. For any candidate label, compute where its internal nonconformity score would rank among the calibration scores in each dimension. Then take the largest of these dimension-wise ranks. The candidate is accepted if its largest rank is not larger than the chosen calibration threshold q . In words, the candidate label for the test instance is accepted if its internal nonconformity score is not too extreme in its worst dimension.

The rank-space **Max-Rank** rule is proven to be valid by Timans et al. (2025). In this paper, however, we distinguish between this rank-space rule and its *score-space Max-Rank implementation*. The latter is the practical translation of the rank condition on candidate labels into dimension-wise thresholds on the internal nonconformity scores. This translation is needed because practical implementations usually output intervals in the real output dimensions.

We argue that the score-space **Max-Rank** implementation is one order statistic too small. It uses the q th smallest calibration score in each dimension, whereas the rank-space rule corresponds to the $(q + 1)$ st smallest calibration score. At the upper boundary of the order statistic the conformal prediction convention is to use an infinite threshold if this next calibration score does not exist. The score-space **Max-Rank** implementation lacks this feature. We refer to these two issues as *the rank-to-score conversion errors*. We note that they do not invalidate the rank-space **Max-Rank** rule itself. Rather, they show that the original implementation in the score space is not a faithful representation of the rank-space rule.

In this paper we propose a straightforward **Corrected Max-Rank** implementation in the score space that uses the $(q + 1)$ st smallest dimension-wise calibration score when it exists, and $+\infty$ otherwise. We prove that this corrected implementation restores the finite-sample validity guarantee under exchangeability and dimension-separable internal scores without any additional correction of the significance level.

The contributions of this paper are as follows.

- We identify two rank-to-score conversion errors in the score-space **Max-Rank** implementation.

- We propose the score-space **Corrected Max-Rank** implementation that uses the next dimension-wise order statistic and the $+\infty$ convention.
- We prove finite-sample validity of the score-space **Corrected Max-Rank** implementation under exchangeability and dimension-separable internal scores.
- We empirically compare the score-space **Max-Rank** implementation with the score-space **Corrected Max-Rank** implementation on synthetic multi-target residual distributions.

The remainder of the paper is organised as follows. Section 2 reviews inductive conformal prediction for multi-target regression and introduces the notation used throughout. Section 3 presents the rank-space **Max-Rank** rule together with its original score-space implementation and identifies the two rank-to-score conversion errors that affect the original score-space implementation. Section 4 introduces **Corrected Max-Rank**, the corrected score-space implementation, and proves that it faithfully represents the rank-space rule and therefore inherits its finite-sample coverage guarantee. Section 5 empirically compares **Max-Rank** and **Corrected Max-Rank** on synthetic multi-target internal score distributions. Section 6 discusses the implications of the correction, and situates **Corrected Max-Rank** within the broader literature on hyperrectangular conformal prediction regions for multi-target regression. Section 7 concludes and outlines directions for future work.

2. Inductive Conformal Prediction

We deal with the problem of uncertainty quantification for multi-target regression. Assume an underlying multi-target regression model $\hat{\mathbf{f}} : \mathcal{X} \rightarrow \mathcal{Y}$ with $\mathcal{X} \subseteq \mathbb{R}^p$ and $\mathcal{Y} \subseteq \mathbb{R}^m$ has already been learned and is used to predict m output variables from p input variables. In addition, we observe a calibration dataset $\mathcal{D}_{cal} = \{(\mathbf{X}^{(i)}, \mathbf{Y}^{(i)})\}_{i=1}^n$ consisting of n labelled instances $(\mathbf{X}^{(i)}, \mathbf{Y}^{(i)})$, where $\mathbf{X}^{(i)} = (X_1^{(i)}, \dots, X_p^{(i)})$ and $\mathbf{Y}^{(i)} = (Y_1^{(i)}, \dots, Y_m^{(i)})$ are random vectors with values from $\mathcal{X}$ and $\mathcal{Y}$, respectively. Given a new test instance $\mathbf{X}^{(n+1)}$, the goal is to quantify the uncertainty of prediction $\hat{\mathbf{Y}}^{(n+1)} = \hat{\mathbf{f}}(\mathbf{X}^{(n+1)})$ with respect to the unknown true label $\mathbf{Y}^{(n+1)}$ using the calibration dataset $\mathcal{D}_{cal}$.

Inductive Conformal Prediction (ICP) (Vovk et al., 2022) provides distribution-free uncertainty quantification using prediction regions. ICP guarantees that the prediction region $\Gamma_{\text{ICP}}^\epsilon \subseteq \mathcal{Y}$ for the test instance $\mathbf{X}^{(n+1)}$ satisfies

$$\mathbb{P} \left(\mathbf{Y}^{(n+1)} \in \Gamma_{\text{ICP}}^\epsilon \left(\mathbf{X}^{(n+1)} \right) \right) \geq 1 - \epsilon \quad (1)$$

i.e., the true m -dimensional label $\mathbf{Y}^{(n+1)}$ is included in $\Gamma_{\text{ICP}}^\epsilon$ with probability at least $1 - \epsilon$. The parameter ϵ is the significance level chosen by the user. This validity guarantee holds under the assumption that the instances in the set $\mathcal{D}_{cal} \cup \{(\mathbf{X}^{(n+1)}, \mathbf{Y}^{(n+1)})\}$ are *exchangeable* (Vovk et al., 2022). Formally, this means that for every permutation π of $\{1, \dots, n+1\}$ we have

$$\left((\mathbf{X}^{(i)}, \mathbf{Y}^{(i)}) \right)_{i=1}^{n+1} \stackrel{d}{=} \left((\mathbf{X}^{(\pi(i))}, \mathbf{Y}^{(\pi(i))}) \right)_{i=1}^{n+1}$$

where $\stackrel{d}{=}$ denotes equality in distribution.

To construct the local prediction region $\Gamma_{\text{ICP}}^\epsilon(\mathbf{X}^{(n+1)}) \subseteq \mathcal{Y}$ ICP employs a nonconformity score function (i.e. nonconformity measure) S_{ICP} . It assesses how unusual the pair $(\mathbf{X}^{(i)}, \mathbf{Y}^{(i)})$ is relative to the model $\hat{\mathbf{f}}$. Formally, S_{ICP} is a measurable mapping

$$S_{\text{ICP}} : (\mathcal{F} \times \mathcal{X} \times \mathcal{Y}) \longrightarrow \bar{\mathbb{R}}; \left(\hat{\mathbf{f}}, \mathbf{x}^{(i)}, \mathbf{y}^{(i)} \right) \mapsto S_{\text{ICP}} \left(\hat{\mathbf{f}}, \mathbf{x}^{(i)}, \mathbf{y}^{(i)} \right) \quad (2)$$

where $\mathcal{F}$ is the domain of all models trained on a proper training dataset $\mathcal{D}_{tr}$ that is separate from the calibration dataset $\mathcal{D}_{cal}$ (Vovk et al., 2022; Liang et al., 2025).¹ This definition ensures that the nonconformity score function S_{ICP} is independent of the calibration set. Since $\hat{\mathbf{f}}$ is fixed once it has been trained on $\mathcal{D}_{tr}$, and $\mathcal{D}_{cal}$ is independent of $\mathcal{D}_{tr}$, all probabilistic statements in this paper can be read conditionally on $\mathcal{D}_{tr}$ (Liang et al., 2025). For brevity the model $\hat{\mathbf{f}}$ is implied in the remainder of this paper and we simply write $S_{\text{ICP}}(\mathbf{X}^{(i)}, \mathbf{Y}^{(i)})$ when evaluating the random pair $(\mathbf{X}^{(i)}, \mathbf{Y}^{(i)})$.

Using the nonconformity score function S_{ICP} , ICP computes the nonconformity scores $A_{\text{ICP}}^{(i)} = S_{\text{ICP}}(\mathbf{X}^{(i)}, \mathbf{Y}^{(i)})$ for the n instances in $\mathcal{D}_{cal}$. Then, for a new test instance $\mathbf{X}^{(n+1)}$, it constructs the prediction region

$$\Gamma_{\text{ICP}}^\epsilon(\mathbf{X}^{(n+1)}) = \left\{ \mathbf{y} \in \mathcal{Y} : S_{\text{ICP}}(\mathbf{X}^{(n+1)}, \mathbf{y}) \leq q_{\text{ICP}}^{(1-\epsilon)} \right\} \quad (3)$$

where $q_{\text{ICP}}^{(1-\epsilon)} = Q_{1-\epsilon}(\{A_{\text{ICP}}^{(i)}\}_{i=1}^n)$ is the $(1 - \epsilon)$ -quantile of the calibration nonconformity scores. Thus, $\Gamma_{\text{ICP}}^\epsilon(\mathbf{X}^{(n+1)})$ contains all candidate labels $\mathbf{y}$ whose nonconformity score does not exceed $q_{\text{ICP}}^{(1-\epsilon)}$.

The nonconformity score function S_{ICP} determines the shape of the prediction region. Recent publications (Vovk, 2013; Messoudi et al., 2021; Timans et al., 2025; Schlembach et al., 2025) have treated nonconformity scores that induce axis-aligned hyperrectangles. The appeal of this particular shape lies in its interpretability.

3. Max-Rank

Timans et al. (2025) propose **Max-Rank**, an elegant approach that produces hyperrectangular prediction regions and that is computationally efficient. The size of **Max-Rank**'s hyperrectangles is smaller than other methods' (Vovk, 2013; Messoudi et al., 2021; Schlembach et al., 2025) in many situations.²

The rank-space **Max-Rank** rule works in 4 steps:

Step 1: Using an ‘‘internal’’ dimension-wise nonconformity score function $S_{\text{int}} : (\mathcal{F} \times \mathcal{X} \times \mathcal{Y}) \longrightarrow \bar{\mathbb{R}}^m$ calculate the ‘‘internal’’ nonconformity scores $\beta^{(i)} = (\beta_1^{(i)}, \dots, \beta_m^{(i)})$ for all elements of the calibration dataset $\mathcal{D}_{cal}$. Again, for brevity the model $\hat{\mathbf{f}}$ is implied in the remainder of this paper and we simply write $S_{\text{int}}(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})$. A common example of S_{int} is the

1. The definition provided by Vovk et al. (2022) deviates slightly, passing the elements of the proper training dataset $\mathcal{D}_{tr}$ to S_{ICP} instead of the model. Additionally, Vovk's S_{ICP} also receives the test point $\mathbf{X}^{(n+1)}$. This is done under the assumption that S_{ICP} then fits the model to $\mathcal{D}_{tr}$.
2. See Schlembach et al. (2025) for a detailed comparison.

dimension-wise absolute residual.

Step 2: For each output dimension j , determine the rank³

$$\rho_{\text{ICP},j}^{(i)} = \text{rank} \left(\beta_j^{(i)}, \left\{ \beta_j^{(k)} \right\}_{k=1}^n \right) = \left| \left\{ k \in \{1, \dots, n\} : \beta_j^{(k)} \leq \beta_j^{(i)} \right\} \right|. \quad (4)$$

Step 3: For each $i \in \{1, \dots, n\}$ compute the maximum rank $\alpha_{\text{ICP},\text{MR}}^{(i)} = S_{\text{ICP},\text{MR}}(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})$ across all dimensions with

$$S_{\text{ICP},\text{MR}}(\mathbf{x}^{(i)}, \mathbf{y}^{(i)}) = \max_{j=1}^m \left(\rho_{\text{ICP},j}^{(i)} \right) \quad (5)$$

Step 4: For a new test instance $\mathbf{x}^{(n+1)}$ construct the prediction region

$$\Gamma_{\text{ICP},\text{MR}}^\epsilon \left(\mathbf{x}^{(n+1)} \right) = \left\{ \mathbf{y} \in \mathcal{Y} : S_{\text{ICP},\text{MR}} \left(\mathbf{x}^{(n+1)}, \mathbf{y} \right) \leq q_{\text{ICP}}^{(1-\epsilon)} \right\}. \quad (6)$$

Equation (6) states an inclusion criterion in the rank space. A label $\mathbf{y}$ is included in the prediction region if its maximum rank does not exceed the $1 - \epsilon$ quantile $q_{\text{ICP}}^{(1-\epsilon)}$ of the maximum ranks of the elements in the calibration set.

Timans et al. (2025) translate this inclusion criterion into the score space by selecting, in each dimension j , the internal nonconformity score of rank q as the dimension-wise threshold (Algorithm 2, Step 8 in their article), obtaining

$$\tilde{\Gamma}_{\text{ICP},\text{MR}}^\epsilon \left(\mathbf{x}^{(n+1)} \right) = \left\{ \mathbf{y} \in \mathcal{Y} : S_{\text{int},j} \left(\mathbf{x}^{(n+1)}, \mathbf{y} \right) \leq \beta_j^{(\pi_j^{(q)})}, \forall j \in \{1, \dots, m\} \right\} \quad (7)$$

where π_j is a permutation of $\{1, \dots, n\}$ that sorts the internal nonconformity scores $\beta_j^{(i)}$ of dimension j in ascending order, and $q = q_{\text{ICP}}^{(1-\epsilon)}$. The score-space inclusion criterion states that a label $\mathbf{y}$ is included in the prediction region if its internal nonconformity score $S_{\text{int},j}(\mathbf{x}^{(n+1)}, \mathbf{y})$ does not exceed the q th smallest internal nonconformity score $\beta_j^{(\pi_j^{(q)})}$ in any dimension j . In practice, using the dimension-wise absolute residuals $S_{\text{int}} = \{|\hat{y}_j^{(i)} - y_j^{(i)}|\}_{j=1}^m$ as the internal nonconformity score function, this definition of the score-space inclusion criterion results in the prediction region

$$\tilde{\Gamma}_{\text{ICP},\text{MR}}^\epsilon \left(\mathbf{x}^{(n+1)} \right) = \bigtimes_{j=1}^m \left[\hat{y}_j^{(n+1)} - \beta_j^{(\pi_j^{(q)})}, \hat{y}_j^{(n+1)} + \beta_j^{(\pi_j^{(q)})} \right], \quad (8)$$

which is the Cartesian product of dimension-wise intervals, each centered at the model's prediction $\hat{y}_j^{(n+1)}$ with half-width equal to the q th smallest internal nonconformity score $\beta_j^{(\pi_j^{(q)})}$ in that dimension. Equation (8) matches the interval construction used in the experiments of Timans et al. (2025).

In Section 3.1 we confirm that the rank-space rule of Equation (6) produces valid prediction regions. Section 3.2 shows that Equations (7) and (8) are not faithful implementations of this rank-space rule in the score space. We use $\sim$ to mark the unfaithful score-space implementations, indicating that $\Gamma_{\text{ICP},\text{MR}}^\epsilon \neq \tilde{\Gamma}_{\text{ICP},\text{MR}}^\epsilon$. Section 4 then defines the corrected score-space implementation **Corrected Max-Rank** and proves that its prediction region $\Gamma_{\text{ICP},\text{CMR}}^\epsilon$ coincides with the rank-space region $\Gamma_{\text{ICP},\text{MR}}^\epsilon$ almost surely.

3. We use $R_{\text{ICP},j}^{(i)}$ to denote the corresponding random variable obtained by replacing $\beta_j^{(i)}$ with $B_j^{(i)}$.

3.1. Rank-Space Max-Rank Rule Validity

This section confirms that the rank-space **Max-Rank** rule of Equation (6) produces valid prediction regions and requires no additional correction of the significance level. Timans et al. (2025) already prove this result by directly addressing the question of exchangeability of the maximum rank scores in Appendix A.4 of their article. We provide an alternative proof that approaches validity from the inductive conformal prediction perspective, formalized in Theorem 2 and Corollary 3 below. Readers primarily interested in the practical implementation of **Max-Rank** and its correction may safely skip to Section 3.2.

The motivation for providing an alternative proof of an established result comes from the observation that **Max-Rank**'s maximum rank score of Equation (5) is not a nonconformity score function in the sense of Equation (2). According to Equation (2), the nonconformity score of a pair $(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})$ may depend only on the underlying model $\hat{\mathbf{f}}$ and the pair itself. Computing the ranks in Equation (5), however, requires access to all other elements of the calibration set. The validity guarantee of Equation (1) therefore does not apply to **Max-Rank** directly. Inspired by Liang et al. (2025) we resolve this conflict in three steps. We first introduce *transductive conformal prediction* (TCP), a setting in which the nonconformity score function receives the entire bag of calibration and test data as an additional argument, so that rank computations are permitted. We then reformulate **Max-Rank** as a nonconformity score function for TCP with a fixed underlying model and observe that the general TCP validity argument applies. We also observe that the resulting prediction region is intractable to compute because its threshold depends on the candidate label. Finally, Theorem 2 shows that the tractable ICP prediction region of Equation (6) contains the valid TCP region, so the ICP rank-space **Max-Rank** rule inherits the coverage guarantee in Corollary 3.

In the TCP setting, the nonconformity score function S_{TCP} is defined as a measurable mapping

$$S_{\text{TCP}} : \left(\mathcal{F} \times (\mathcal{X} \times \mathcal{Y})^{(*)} \times \mathcal{X} \times \mathcal{Y} \right) \longrightarrow \bar{\mathbb{R}}, \quad (9)$$

such that

$$\left(\hat{\mathbf{f}}, \mathcal{D} \cup \left\{ \left(\mathbf{x}^{(n+1)}, \mathbf{y} \right) \right\}, \mathbf{x}^{(i)}, \mathbf{y}^{(i)} \right) \mapsto S_{\text{TCP}} \left(\hat{\mathbf{f}}, \mathcal{D} \cup \left\{ \left(\mathbf{x}^{(n+1)}, \mathbf{y} \right) \right\}, \mathbf{x}^{(i)}, \mathbf{y}^{(i)} \right) \quad (10)$$

if $i \in \{1, \dots, n\}$, and

$$\left(\hat{\mathbf{f}}, \mathcal{D} \cup \left\{ \left(\mathbf{x}^{(n+1)}, \mathbf{y} \right) \right\}, \mathbf{x}^{(n+1)}, \mathbf{y} \right) \mapsto S_{\text{TCP}} \left(\hat{\mathbf{f}}, \mathcal{D} \cup \left\{ \left(\mathbf{x}^{(n+1)}, \mathbf{y} \right) \right\}, \mathbf{x}^{(n+1)}, \mathbf{y} \right) \quad (11)$$

for the new test point and its hypothetical label. $(\mathcal{X} \times \mathcal{Y})^{(*)}$ is the set of all bags of elements of $\mathcal{X} \times \mathcal{Y}$ (Vovk et al., 2022) and $\mathcal{D}$ is one such bag of size n .

Remark 1 (TCP with a frozen model) *The interpretation and the definition of S_{TCP} deviate slightly from the one provided by Vovk et al. (2022). In the classical transductive setting, the bag $\mathcal{D} \cup \{(\mathbf{X}^{(n+1)}, \mathbf{y})\}$ is used to train the model, so the model is refit for every hypothetical label $\mathbf{y}$. Here, instead, S_{TCP} receives a fixed model $\hat{\mathbf{f}}$ that has already been trained on data other than $\mathcal{D} \cup \{(\mathbf{X}^{(n+1)}, \mathbf{y})\}$, and the bag enters the nonconformity score only through the ranks. Equivalence with the classical definition is achieved by assuming a model training algorithm that ignores the available data and simply returns the pretrained model $\hat{\mathbf{f}}$ (cf. the remark following Eq. (2) in Liang et al., 2025).*

The prediction region for TCP is defined in the same way as for ICP in Equation (3), i.e.

$$\Gamma_{\text{TCP}}^\epsilon(\mathbf{X}^{(n+1)}) = \left\{ \mathbf{y} \in \mathcal{Y} : S_{\text{TCP}}(\mathbf{X}^{(n+1)}, \mathbf{y}) \leq q_{\text{TCP}}^{(1-\epsilon)} \right\}. \quad (12)$$

Note however that $q_{\text{TCP}}^{(1-\epsilon)}$ now depends on the pair $(\mathbf{X}^{(n+1)}, \mathbf{y})$, rendering the computation of TCP prediction regions intractable in general.

If the elements in $\mathcal{D} \cup \{(\mathbf{X}^{(n+1)}, \mathbf{y})\}$ are exchangeable (Vovk et al., 2022) and S_{TCP} treats them symmetrically (Barber et al., 2023), then TCP produces valid prediction regions according to Equation (1).

In the TCP setting, we can devise a nonconformity score function $S_{\text{TCP}, \text{MR}}$ in the spirit of Max-Rank, i.e.

$$S_{\text{TCP}, \text{MR}}(\hat{\mathbf{f}}, \mathcal{D} \cup \{(\mathbf{x}^{(n+1)}, \mathbf{y})\}, \mathbf{x}^{(i)}, \mathbf{y}^{(i)}) = \max_{j=1}^m \left(\rho_{\text{TCP}, j}^{(i)} \right) \quad (13)$$

such that

$$\alpha_{\text{TCP}, \text{MR}}^{(i)} = \begin{cases} S_{\text{TCP}, \text{MR}}(\hat{\mathbf{f}}, \mathcal{D} \cup \{(\mathbf{x}^{(n+1)}, \mathbf{y})\}, \mathbf{x}^{(i)}, \mathbf{y}^{(i)}) & \text{if } i \in \{1, \dots, n\} \\ S_{\text{TCP}, \text{MR}}(\hat{\mathbf{f}}, \mathcal{D} \cup \{(\mathbf{x}^{(n+1)}, \mathbf{y})\}, \mathbf{x}^{(i)}, \mathbf{y}) & \text{if } i = n + 1. \end{cases} \quad (14)$$

with

$$\rho_{\text{TCP}, j}^{(i)} = \text{rank} \left(\beta_j^{(i)}, \left\{ \beta_j^{(k)} \right\}_{k=1}^{n+1} \right) \quad (15)$$

and

$$\beta^{(i)} = \begin{cases} S_{\text{int}}(\hat{\mathbf{f}}, \mathbf{x}^{(i)}, \mathbf{y}^{(i)}) & \text{if } i \in \{1, \dots, n\} \\ S_{\text{int}}(\hat{\mathbf{f}}, \mathbf{x}^{(i)}, \mathbf{y}) & \text{if } i = n + 1. \end{cases} \quad (16)$$

$S_{\text{TCP}, \text{MR}}$ induces the prediction region

$$\Gamma_{\text{TCP}, \text{MR}}^\epsilon(\mathbf{X}^{(n+1)}) = \left\{ \mathbf{y} \in \mathcal{Y} : S_{\text{TCP}, \text{MR}}(\mathbf{X}^{(n+1)}, \mathbf{y}) \leq q_{\text{TCP}}^{(1-\epsilon)} \right\} \quad (17)$$

that is valid simply by $S_{\text{TCP}, \text{MR}}$ fulfilling the requirements placed on any nonconformity score function S_{TCP} used in the TCP setting. However, as noted above, it is intractable to compute $\Gamma_{\text{TCP}, \text{MR}}^\epsilon$.

The ICP variant of the rank-space Max-Rank rule provides a valid and computationally efficient solution. To prove this, Theorem 2 first investigates the relationship between the prediction regions of the rank-space Max-Rank rule in the ICP and TCP cases. Its proof is given in Appendix A and relies on Lemmas 10 and 11 which establish the relation between the ICP and TCP Max-Rank scores.

Theorem 2 (Containment of ICP Rank-Space Max-Rank Rule) *Using the same internal nonconformity score function S_{int} and model $\hat{\mathbf{f}}$, the prediction region $\Gamma_{\text{ICP}, \text{MR}}^\epsilon$ defined in Equation (6) satisfies*

$$\Gamma_{\text{ICP}, \text{MR}}^\epsilon(\mathbf{X}^{(n+1)}) \supseteq \Gamma_{\text{TCP}, \text{MR}}^\epsilon(\mathbf{X}^{(n+1)}) \quad (18)$$

with $\Gamma_{\text{TCP}, \text{MR}}^\epsilon$ as defined in Equation (17), and assuming that the internal nonconformity scores $\{B_j^{(i)}\}_{i=1}^{n+1}$ are almost surely distinct for each j .

Corollary 3 then follows from the validity of the TCP variant of the rank-space **Max-Rank** rule. Its proof is also given in Appendix A.

Corollary 3 (Validity of ICP Rank-Space Max-Rank Rule) *Under the assumptions of Theorem 2, and assuming further that the elements in $\mathcal{D}_{cal} \cup \{(\mathbf{X}^{(n+1)}, \mathbf{Y}^{(n+1)})\}$ are exchangeable, the prediction region $\Gamma_{ICP,MR}^\epsilon$ as defined in Equation (6) satisfies*

$$\mathbb{P}\left(\mathbf{Y}^{(n+1)} \in \Gamma_{ICP,MR}^\epsilon\left(\mathbf{X}^{(n+1)}\right)\right) \geq 1 - \epsilon. \quad (19)$$

3.2. Max-Rank Rank-to-Score Conversion Errors

The rank-space **Max-Rank** rule has a number of benefits compared to other methods producing axis-aligned hyperrectangular prediction regions that we discuss in Section 6. Despite these benefits, we have identified two errors with the score-space **Max-Rank** implementation for ICP as proposed by Timans et al. (2025) that make this implementation invalid.

Error 1: Lack of infinite prediction regions For $\epsilon < 1/(n+1)$ the quantile function in *Step 4* will return an infinite value. This infinite $q_{ICP}^{(1-\epsilon)}$ is not treated in the initial formulation of **Max-Rank** which is problematic because there is no $\beta_j^{(i)}$ with infinite rank in any dimension.

Error 2: Overoptimistic intervals Building the prediction region according to Equation (8), i.e. by adding and subtracting $\beta_j^{(\pi_j^{(q)})}$ with $q = q_{ICP}^{(1-\epsilon)}$ from $\hat{\mathbf{y}}^{(n+1)}$ in each dimension j is overly optimistic. An example for $n = 5$ and $m = 1$ illustrates why. Let $\{\beta_1^{(i)}\}_{i=1}^5 = \{0.1, 0.2, 0.3, 0.4, 0.5\}$ be the set of internal nonconformity scores and $\{\alpha_{ICP,MR}^{(i)}\}_{i=1}^5 = \{1, 2, 3, 4, 5\}$ their associated maximum ranks. For $\epsilon = 0.35$ the quantile function returns the rank $q_{ICP}^{(1-\epsilon)} = 4$, hence the prediction region for $\mathbf{x}^{(n+1)}$ is $[\hat{y}^{(n+1)} - 0.4, \hat{y}^{(n+1)} + 0.4]$.

Let us now consider the pair $(\mathbf{x}^{(n+1)}, y)$ with a hypothetical label y such that $\beta_1^{(n+1)} = |\hat{y}^{(n+1)} - y| = 0.45$. This hypothetical label falls outside the score-space prediction region $\tilde{\Gamma}_{ICP,MR}^\epsilon(\mathbf{x}^{(n+1)})$ generated by **Max-Rank**. At the same time, according to Equation (5), the maximum rank of the pair $(\mathbf{x}^{(n+1)}, y)$ is $\alpha_{ICP,MR}^{(n+1)} = 4$. Interpreting the $\alpha_{ICP,MR}^{(i)}$ as nonconformity scores and using Equation (3) as a reference for the construction of a prediction region, the hypothetical label y should have been included in the prediction region because $\alpha_{ICP,MR}^{(n+1)} = 4 \leq q_{ICP}^{(1-\epsilon)}$.

Thus, Error 2 is due to the use of the q th smallest calibration score in each dimension, whereas the rank-space rule corresponds to the $(q+1)$ st smallest calibration score.

4. Score-Space Corrected Max-Rank Implementation

The previous section identifies two rank-to-score conversion errors in the score-space **Max-Rank** implementation. In this section we define the corrected score-space implementation, denoted by **Corrected Max-Rank**. Unlike $\tilde{\Gamma}_{ICP,MR}^\epsilon$ in Equations (7) and (8), it represents the valid rank-space **Max-Rank** rule faithfully by dimension-wise thresholds on the internal nonconformity scores, as established by Theorem 5 below. We therefore denote **Corrected Max-Rank**'s prediction region by the symbol $\Gamma_{ICP,CMR}^\epsilon$ without tilde.

The prediction region of the **Corrected Max-Rank** implementation in the score space, $\Gamma_{\text{ICP,CMR}}^\epsilon$, is

$$\Gamma_{\text{ICP,CMR}}^\epsilon(\mathbf{x}^{(n+1)}) = \left\{ \mathbf{y} \in \mathcal{Y} : S_{\text{int},j}(\mathbf{x}^{(n+1)}, \mathbf{y}) \leq \beta_j^{(\pi_j^{(q+1)})}, \forall j \in \{1, \dots, m\} \right\} \quad (20)$$

with $\beta_j^{(\pi_j^{(q+1)})} = +\infty$ if $q+1 > n$, and $q = q_{\text{ICP}}^{(1-\epsilon)}$. The **Corrected Max-Rank** score-space inclusion criterion states that a label $\mathbf{y}$ is included in the prediction region if its internal nonconformity score $S_{\text{int},j}(\mathbf{x}^{(n+1)}, \mathbf{y})$ does not exceed the $(q+1)$ st smallest internal nonconformity score $\beta_j^{(\pi_j^{(q+1)})}$ in any dimension j . Using the dimension-wise absolute residuals $S_{\text{int}} = \{|\hat{y}_j^{(i)} - y_j^{(i)}|\}_{j=1}^m$ as the internal nonconformity score function, this results in the prediction region

$$\Gamma_{\text{ICP,CMR}}^\epsilon(\mathbf{x}^{(n+1)}) = \bigtimes_{j=1}^m \left[\hat{y}_j^{(n+1)} - \beta_j^{(\pi_j^{(q+1)})}, \hat{y}_j^{(n+1)} + \beta_j^{(\pi_j^{(q+1)})} \right] \quad (21)$$

which is the Cartesian product of dimension-wise intervals, each centered at the model's prediction $\hat{y}_j^{(n+1)}$ now with half-width equal to the $(q+1)$ st smallest internal nonconformity score $\beta_j^{(\pi_j^{(q+1)})}$ in that dimension.

To overcome Error 1, we explicitly handle the case where the quantile function returns a maximum rank that does not exist among the n calibration scores. Whenever $q_{\text{ICP}}^{(1-\epsilon)} = +\infty$, or equivalently when $\lceil (1-\epsilon)(n+1) \rceil > n$, the corrected score-space implementation returns prediction intervals of infinite size in all dimensions.

To overcome Error 2, the corrected implementation uses the $q_{\text{ICP}}^{(1-\epsilon)} + 1$ smallest internal calibration score in each dimension instead of the $q_{\text{ICP}}^{(1-\epsilon)}$ smallest. If this $q_{\text{ICP}}^{(1-\epsilon)} + 1$ smallest score does not exist because $q_{\text{ICP}}^{(1-\epsilon)} + 1 > n$, we again use an infinite threshold.

We now show that this corrected conversion gives the correct score-space representation of the rank-space **Max-Rank** rule as defined in Equation (6). Theorem 5 shows this. For its proof we employ Assumption 1 and Lemma 4.

Assumption 1 *For each $j \in \{1, \dots, m\}$, the component $S_{\text{int},j}$ of the internal nonconformity score function depends on the label $\mathbf{y}$ only through its j th dimension, i.e.*

$$S_{\text{int},j}(\hat{\mathbf{f}}, \mathbf{x}, \mathbf{y}) = S_{\text{int},j}(\hat{\mathbf{f}}, \mathbf{x}, \mathbf{y}') \quad \forall \mathbf{y}, \mathbf{y}' \in \mathcal{Y} \text{ with } y_j = y'_j. \quad (22)$$

This assumption ensures that the internal nonconformity score function S_{int} is dimension-separable. Absolute residuals, normalised residuals (Papadopoulos et al., 2008), and CQR (Romano et al., 2019) scores all satisfy this. Under Assumption 1 we write, with slight abuse of notation, $S_{\text{int},j}(\hat{\mathbf{f}}, \mathbf{x}, y_j)$ for the common value of $S_{\text{int},j}(\hat{\mathbf{f}}, \mathbf{x}, \mathbf{y})$ over all labels $\mathbf{y}$ sharing the j th component y_j . Assumption 1 assures that *Step 4* in the original formulation of the score-space **Max-Rank** implementation can be extended to nonconformity scores other than just the absolute residuals.

Lemma 4 Let $\{\beta^{(i)}\}_{i=1}^n$ be n almost surely distinct real numbers, and let π be a permutation of $\{1, \dots, n\}$ such that $\beta^{(\pi(1))} < \beta^{(\pi(2))} < \dots < \beta^{(\pi(n))}$. We define $\text{rank}(v) = |\{k : \beta^{(k)} \leq v\}|$ for any $v \in \mathbb{R}$, and adopt the convention $\beta^{(\pi(l))} = +\infty$ for any $l \geq n+1$. Then for every $q \in \{0, 1, \dots, n, n+1, +\infty\}$,

$$\text{rank}(v) \leq q \iff v < \beta^{(\pi(q+1))}. \quad (23)$$

Proof For $q \in \{0, 1, \dots, n\}$

($\Rightarrow$) If $\text{rank}(v) \leq q$, then at most q of the $\beta^{(k)} \in (-\infty, v]$. In particular $\beta^{(\pi(q+1))} \notin (-\infty, v]$, i.e. $\beta^{(\pi(q+1))} > v$.

($\Leftarrow$) If $v < \beta^{(\pi(q+1))}$, then by strict monotonicity of the sorted sequence, there are at most q elements $\beta^{(\pi(k))}$ such that $\beta^{(\pi(k))} \leq v < \beta^{(\pi(q+1))}$. This is because the elements that satisfy $\beta^{(\pi(k))} \leq v$ need to be among the q smallest ones.

For $q \in \{n+1, +\infty\}$ both sides hold trivially for any $v \in \mathbb{R}$. The left side holds because $\text{rank}(v) \leq n$ by definition. The right side holds because $q+1 \geq n+1$ and therefore $\beta^{(\pi(q+1))} = +\infty$. $\blacksquare$

Theorem 5 If $q = q_{ICP}^{(1-\epsilon)}$, Assumption 1 holds, and the internal nonconformity scores $\{\beta_j^{(i)}\}_{i=1}^n$ are almost surely distinct for each $j \in \{1, \dots, m\}$, then

$$\Gamma_{ICP, MR}^\epsilon(\mathbf{x}^{(n+1)}) = \Gamma_{ICP, CMR}^\epsilon(\mathbf{x}^{(n+1)}) \quad \text{almost surely,} \quad (24)$$

i.e. **Corrected Max-Rank** as defined in Equation (20) is the correct score-space implementation of the rank-space prediction region defined in Equation (6).

Proof By definition of $S_{ICP, MR}$ in Equation (5) and Equation (6)

$$\Gamma_{ICP, MR}^\epsilon = \left\{ \mathbf{y} : \max_{j=1}^m \text{rank} \left(S_{int, j}(\hat{\mathbf{f}}, \mathbf{X}^{(n+1)}, y_j), \{\beta_j^{(k)}\}_{k=1}^n \right) \leq q \right\} \quad (25)$$

The maximum rank is $\leq q$ if and only if each dimension's rank is $\leq q$. Applying Lemma 4 to each dimension, we rewrite the set as

$$\Gamma_{ICP, MR}^\epsilon = \bigcap_{j=1}^m \left\{ \mathbf{y} \in \mathcal{Y} : S_{int, j}(\hat{\mathbf{f}}, \mathbf{X}^{(n+1)}, y_j) < \beta_j^{(\pi_j^{(q+1)})} \right\}. \quad (26)$$

Under Assumption 1, the j th constraint depends only on y_j , so the intersection equals the Cartesian product.

Due to the strict inequality (26) $\beta_j^{(\pi_j^{(q+1)})}$ should not be included itself. The difference is exactly the boundary set

$$\partial = \bigcup_{j=1}^m \left\{ \mathbf{y} : S_{int, j}(\hat{\mathbf{f}}, \mathbf{X}^{(n+1)}, y_j) = \beta_j^{(\pi_j^{(q+1)})} \right\}, \quad (27)$$

on which the rank equals $q+1$ rather than q . Under the almost-sure distinctness assumption, ∂ has probability zero under any continuous data distribution, hence $\Gamma_{ICP, MR}^\epsilon(\mathbf{x}^{(n+1)})$ and $\Gamma_{ICP, CMR}^\epsilon(\mathbf{x}^{(n+1)})$ coincide almost surely. $\blacksquare$

Theorem 5 proves that **Corrected Max-Rank** is the correct score-space implementation of the rank-space **Max-Rank** rule. Thus, from the validity of **Max-Rank** in the rank space it follows that **Corrected Max-Rank** produces valid prediction regions according to Equation (1).

The equality in Theorem 5 holds almost surely rather than pointwise. Lemma 4 shows that the rank-space rule corresponds to strict dimension-wise thresholds, whereas **Corrected Max-Rank** uses closed intervals and therefore the prediction region additionally includes the boundary set ∂ from Equation (27), which carries no probability mass under any continuous data distribution. Because $\Gamma_{\text{ICP},\text{MR}}^\epsilon(\mathbf{x}^{(n+1)}) \subseteq \Gamma_{\text{ICP},\text{CMR}}^\epsilon(\mathbf{x}^{(n+1)})$ always holds, **Corrected Max-Rank** errs, if at all, on the conservative side and remains valid. This distinction is the reason we keep separate symbols for the rank-space rule and its score-space implementation.

Corollary 6 *Under the assumptions of Theorem 5, using the dimension-wise absolute residuals $\mathbf{S}_{\text{int}} = \{|\hat{y}_j^{(i)} - y_j^{(i)}|\}_{j=1}^m$ as the internal nonconformity score function, the rank-space **Max-Rank** rule of Equation (6) is implemented in the score space by the hyperrectangle*

$$\Gamma_{\text{ICP},\text{MR}}^\epsilon(\mathbf{x}^{(n+1)}) = \Gamma_{\text{ICP},\text{CMR}}^\epsilon(\mathbf{x}^{(n+1)}) = \bigtimes_{j=1}^m \left[\hat{y}_j^{(n+1)} - \beta_j^{(\pi_j^{(q+1)})}, \hat{y}_j^{(n+1)} + \beta_j^{(\pi_j^{(q+1)})} \right] \quad (28)$$

with the convention $\beta_j^{(\pi_j^{(l)})} = +\infty$ for $l \geq n+1$, where the equalities hold almost surely.

Proof The absolute residuals satisfy Assumption 1 with $\mathbf{S}_{\text{int},j}(\hat{\mathbf{f}}, \mathbf{x}^{(n+1)}, y_j) = |\hat{y}_j^{(n+1)} - y_j|$. By Theorem 5, a label $\mathbf{y}$ belongs to $\Gamma_{\text{ICP},\text{MR}}^\epsilon(\mathbf{x}^{(n+1)})$ if and only if $|\hat{y}_j^{(n+1)} - y_j| \leq \beta_j^{(\pi_j^{(q+1)})}$ for all $j \in \{1, \dots, m\}$. Solving each dimension-wise criterion for y_j yields the interval $[\hat{y}_j^{(n+1)} - \beta_j^{(\pi_j^{(q+1)})}, \hat{y}_j^{(n+1)} + \beta_j^{(\pi_j^{(q+1)})}]$, and the region is the Cartesian product of these intervals, recovering Equation (21). $\blacksquare$

Listing 1 provides an exemplary `python3` implementation of the corrected dimension-wise thresholds. The ranks are computed as shown in Equation (4). Because `numpy`'s indices start at 0, the $(q+1)$ st smallest internal nonconformity score sits at index q .

The experimental results in Section 5 support the argument for the identified errors as well as the proposed solution.

5. Experiments

To validate the theoretical results described in Section 3, we compare the score-space **Max-Rank** implementation from Equation (8) and the score-space implementation of **Corrected Max-Rank** from Equation (21) empirically on synthetic data. The two methods share the same implementation with the only difference between them being the solution proposed to Error 2. Both methods implement the solution to Error 1. This is necessary to avoid crashes of the program running the experiments. Thus the empirical comparison isolates the effect of the solution to Error 2.

To simplify the data generation process, we omit the training of an underlying model, instead sampling values for $(\hat{\mathbf{y}}^{(i)} - \mathbf{y}^{(i)}) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\sigma})$ from an m -dimensional multivariate

Listing 1: Example python3 implementation of the Corrected Max-Rank dimension-wise thresholds. For absolute residuals the prediction region is the hyperrectangle of Corollary 6 with the returned thresholds as half-widths of the sides.

```
import numpy as np

def corrected_max_rank_thresholds(scores, epsilon):
    # scores: internal nonconformity scores of the calibration set
    n, m = scores.shape
    k = int(np.ceil((1 - epsilon) * (n + 1)))

    # Error 1 fix: the conformal quantile is infinite
    if k > n:
        return np.full(m, np.inf)

    # + 1 so ranks don't start at 0
    ranks = np.argsort(np.argsort(scores, axis=0), axis=0) + 1
    q = np.sort(ranks.max(axis=1))[k - 1] # rank-space threshold

    # Error 1 fix: the (q + 1)st smallest score does not exist
    if q + 1 > n:
        return np.full(m, np.inf)

    # Error 2 fix: (q + 1)st smallest score is at index q
    return np.sort(scores, axis=0)[q]
```

normal distribution, where σ is an $m \times m$ matrix with values equalling 1 on the main diagonal and σ otherwise. The absolute residuals serve as the “internal” nonconformity score function, such that $\beta_j^{(i)} = |\hat{y}_j^{(i)} - y_j^{(i)}|$.

We conduct three sets of experiments.

Set 1 varies the correlation coefficient across the dimensions j of the $\beta_j^{(i)}$ with $\sigma \in \{0, 0.05, 0.10, \dots, 1.0\}$ while the number of elements in the calibration set $n = 200$ and the number of dimensions of the label space $m = 4$ remain fixed.

Set 2 varies the number of dimensions of the label space $m \in \{2, 3, \dots, 12\}$ while $n = 200$ and $\sigma = 0.5$ remain constant.

Set 3 varies the number of elements in the calibration set $n \in \{10, 25, 50, 100, 200, 400, 800, 1600, 3200, 6400\}$ while $m = 4$ and $\sigma = 0.5$ remain fixed.

The test set is of the same size n as the calibration set, generated independently from the latter for all experiments.

The score-space Max-Rank and Corrected Max-Rank implementations are evaluated by measuring the error rate, i.e. the fraction of test points that are not contained within the prediction region, and the mean interval width, i.e. the average side length of the hyperrectangular prediction region. All experiments are repeated 1000 times and the results are averaged.

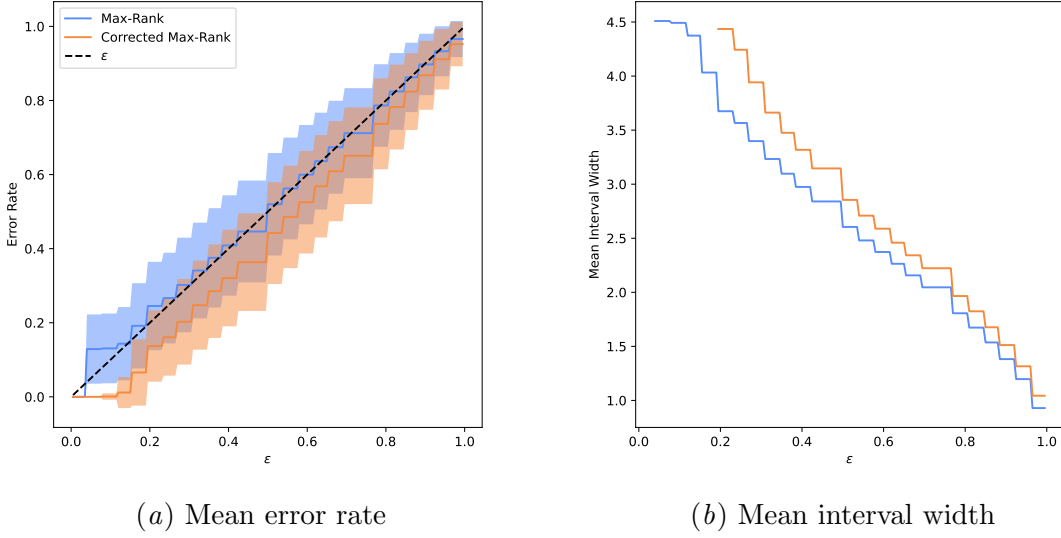


Figure 1: Mean error rate and mean interval width for the score-space **Max-Rank** and **Corrected Max-Rank** implementations with $n = 25$, $m = 4$, and a correlation coefficient of $\sigma = 0.5$. The shaded area represents $\pm$ one standard deviation.

Figure 1 shows these metrics for the case $n = 25$, $m = 4$, and $\sigma = 0.5$. The shaded area indicates the standard deviation computed over the 1000 repetitions of the experiment. We observe the overoptimistic approach of the score-space **Max-Rank** implementation in Figure 1(a), where the measured error rate exceeds the significance level ϵ on multiple occasions, especially for smaller values of ϵ . The score-space **Corrected Max-Rank** implementation remains valid for any $\epsilon \in [0.005, 0.995]$ in this example. In Figure 1(b) we observe that the mean interval width for **Corrected Max-Rank** is larger than **Max-Rank**'s. This difference is more pronounced for smaller values of ϵ .

To compare the score-space **Max-Rank** and **Corrected Max-Rank** implementations over the range of a varying parameter we use two additional metrics. The *mean excess error rate* indicates the average difference of the measured error rate and ϵ throughout the entire range of ϵ . A positive value shows that the method is not valid on average while a negative value supports the claim of validity. The *mean relative interval width difference* is the mean over all values of $\epsilon \geq (m + 1)/(n + 1)$ of the relative difference of the mean interval width between the two methods. The condition $\epsilon \geq (m + 1)/(n + 1)$ is necessary to exclude infinite intervals.⁴ This metric indicates by what fraction the mean side lengths of the **Corrected**

4. The maximum rank of any element in $\mathcal{D}_{cal}$ is at most n , and under the almost-sure distinctness of $\{B_j^{(i)}\}_{i=1}^n$ in each dimension exactly one element attains rank n in dimension j . Hence at most m elements in $\mathcal{D}_{cal}$ can have maximum rank equal to n . Since $q_{ICP}^{(1-\epsilon)}$ is the $\lceil (1-\epsilon)(n+1) \rceil$ -th order statistic of the maximum ranks, finite intervals require $q_{ICP}^{(1-\epsilon)} \leq n-1$, which in the worst case (the top m order statistics all equal to n) demands $\lceil (1-\epsilon)(n+1) \rceil \leq n-m$, i.e. $\epsilon \geq (m+1)/(n+1)$.

Max-Rank hyperrectangles are larger than the **Max-Rank** ones. In all experiments results were collected for $\epsilon \in \{0.005, 0.010, \dots, 0.995\}$.

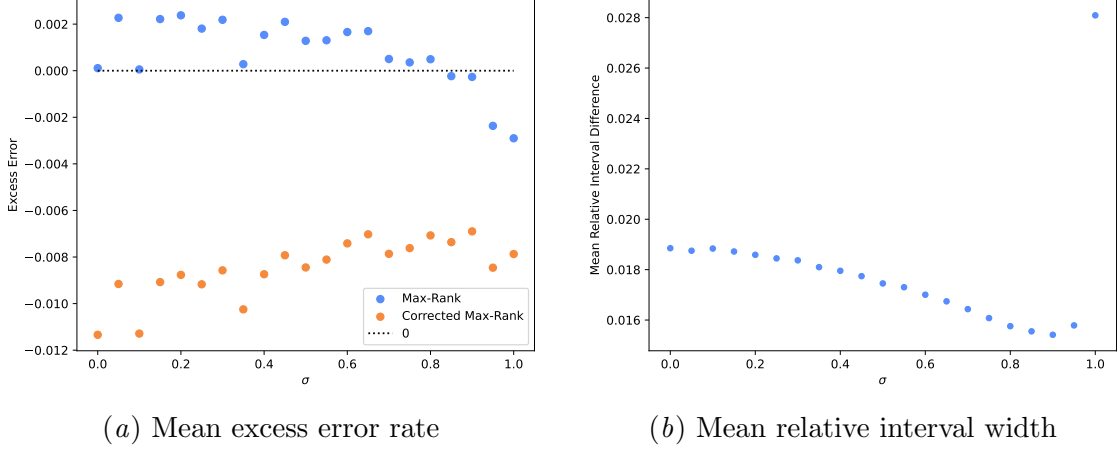


Figure 2: Mean excess error rate and mean relative interval width for the score-space **Max-Rank** and **Corrected Max-Rank** implementations with $n = 200$, $m = 4$, and a correlation coefficient $\sigma \in \{0, 0.05, 0.10, \dots, 1.0\}$.

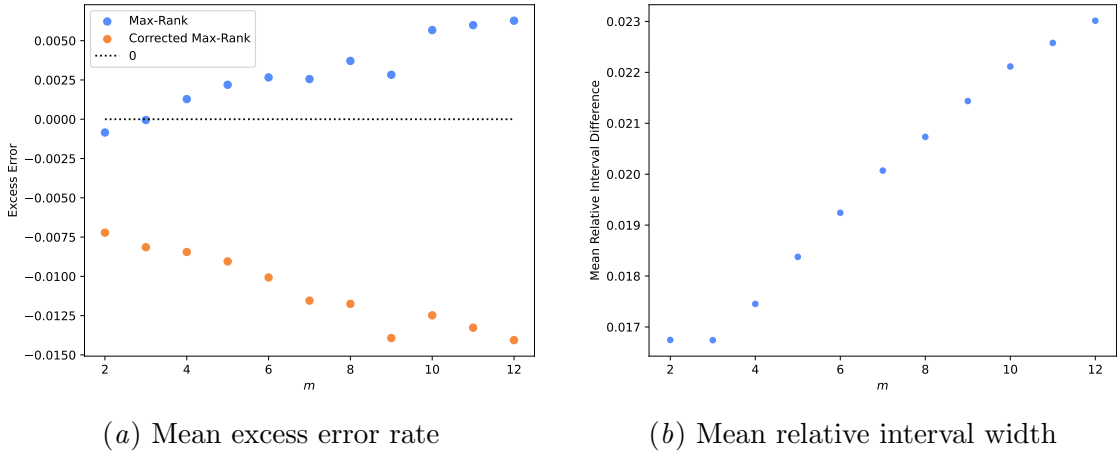


Figure 3: Mean excess error rate and mean relative interval width for the score-space **Max-Rank** and **Corrected Max-Rank** implementations with $n = 200$, $\sigma = 0.5$, and a number of dimensions $m \in \{2, 3, \dots, 12\}$.

Figure 2 displays the results for *Set 1*, comparing the score-space **Max-Rank** and **Corrected Max-Rank** implementations for an increasing correlation σ between the dimensions j of the internal nonconformity scores $\beta_j^{(i)}$. They show that the excess error rate of the **Max-Rank**

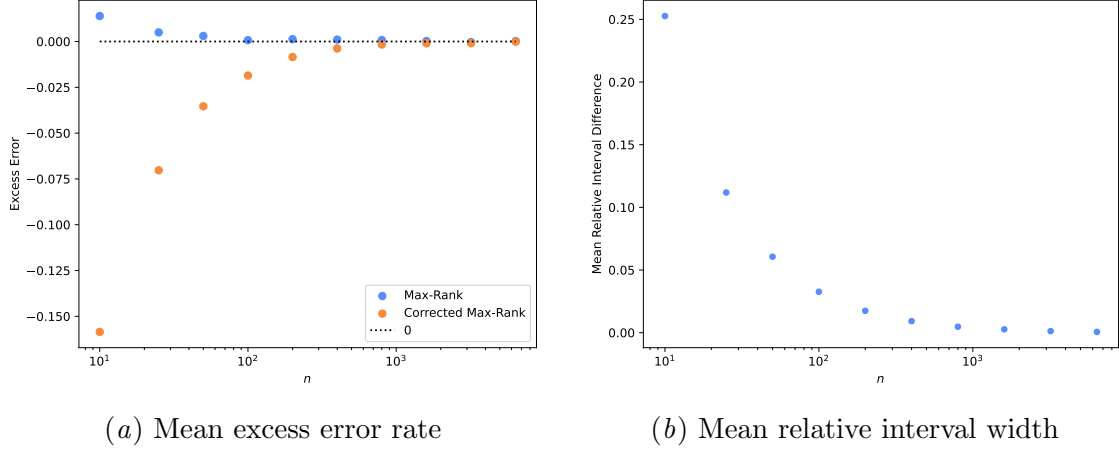


Figure 4: Mean excess error rate and mean relative interval width for the score-space **Max-Rank** and **Corrected Max-Rank** implementations with $m = 4$, $\sigma = 0.5$ and a number of samples in the calibration set $n \in \{10, 25, 50, 100, 200, 400, 800, 1600, 3200, 6400\}$.

implementation is largest for uncorrelated $\beta_j^{(i)}$ and decreases with an increase in σ . The relative difference in the mean interval width of the prediction regions between the **Max-Rank** and **Corrected Max-Rank** implementations first decreases slightly with an increase in σ , until $\sigma = 0.9$. For $\sigma \geq 0.9$ the mean relative interval width difference suddenly grows with increasing σ , reaching its largest value for $\sigma = 1$. The largest prediction regions always stem from the score-space **Corrected Max-Rank** implementation.

Figure 3 shows the results for *Set 2*, comparing the score-space **Max-Rank** and **Corrected Max-Rank** implementations for an increasing number of dimensions m . We observe that the excess error rate of the **Max-Rank** implementation increases with the number of dimensions. The difference in mean interval width between the **Max-Rank** and **Corrected Max-Rank** implementations also increases with m , with the **Corrected Max-Rank** implementation always producing larger prediction regions.

The experiments in *Set 3* show that the excess error rate of the score-space **Max-Rank** implementation decreases with an increase in the number of elements in the calibration set n . The **Corrected Max-Rank** implementation produces prediction regions with a larger mean interval width than the **Max-Rank** implementation, with the difference being largest for small n and decreasing with an increase in n . For $n = 1600$ the difference in mean interval width is almost imperceptible. Figure 4 shows the corresponding results.

Across all experiments we observe that the difference in the mean interval width between the two methods generally moves in the same direction as the excess error rate of **Max-Rank**. **Corrected Max-Rank** is conservatively valid in all experiments.

6. Discussion and Related Work

Validity and Score-Space Implementation A central point of this paper is the distinction between **Max-Rank** as a rank-space rule and **Max-Rank** as a dimension-wise score-space implementation. At the rank level, **Max-Rank** accepts a candidate label $\mathbf{y}$ whenever its maximum rank does not exceed the calibration threshold $q_{\text{ICP}}^{(1-\epsilon)}$. This rule is justified by comparison with the corresponding transductive rank-space construction in Theorem 2 and Corollary 3. The transductive construction treats the calibration examples and the candidate test example symmetrically, and therefore inherits the standard conformal validity argument under exchangeability and symmetry. The difficulty arises when the rank-space condition is translated into dimension-wise intervals, as in $\tilde{\Gamma}_{\text{ICP},\text{MR}}^\epsilon$. We therefore show that **Max-Rank** is valid in the rank space but that the original score-space implementation does not represent the rank-space rule faithfully, and that the corrected implementation $\Gamma_{\text{ICP},\text{CMR}}^\epsilon$ does.

The correction is most relevant when the size of the calibration set n is small. This is visible in the experiments, especially for small n , small ϵ , and larger numbers of output dimensions m . Figure 4 illustrates this difference, both for the empirical coverage and the difference in the average side length of the prediction region. As n grows, the difference between $\beta_j^{(\pi_j^{(q)})}$ and $\beta_j^{(\pi_j^{(q+1)})}$ typically becomes smaller. This explains why the empirical difference between the **Max-Rank** and **Corrected Max-Rank** implementations decreases for large calibration sets. Timans et al. (2025) discuss that their original implementation requires a large sample in their Appendix D.1.

Dependence Between Output Dimensions The experiments indicate that the behaviour of **Max-Rank** and **Corrected Max-Rank** depends strongly on the correlation between the dimension-wise internal scores. When the residuals are weakly correlated, extreme ranks tend to occur in different examples across different dimensions. This makes joint coverage more difficult and increases the risk that the original implementation undercovers.

Comparison with Bonferroni Correction Bonferroni CP (Vovk, 2013) controls the joint error across all output dimensions by choosing local significance levels ϵ_j such that $\epsilon_j = \epsilon/m$. This gives a simple and robust validity guarantee, but it can be conservative, especially when the output dimensions are positively correlated. In contrast, **Max-Rank** adapts to the observed dependence structure in the calibration data. The **Corrected Max-Rank** implementation can therefore be viewed as a dependence-aware alternative to Bonferroni CP that produces smaller prediction regions in general, and remains computationally efficient (Timans et al., 2025; Schlembach et al., 2025).

Timans et al. (2025) further claim that **Max-Rank** is equivalent to Bonferroni CP in the worst-case scenario, which would make swapping Bonferroni CP for **Max-Rank** a free improvement for any application already using Bonferroni CP. After our corrections this is no longer the case. The 1D example from the description of Error 2 in Section 3.2 illustrates the difference. Bonferroni CP computes $\epsilon' = \epsilon/m = 0.35/1$ and returns the $\lceil (1 - \epsilon')(5 + 1) \rceil = 4$ th smallest internal nonconformity score $\beta_1^{(\pi_1^{(4)})} = 0.4$ as the threshold in the score space. **Corrected Max-Rank**, in contrast, returns the $\lceil (1 - \epsilon)(5 + 1) \rceil + 1 = 5$ th smallest internal nonconformity score $\beta_1^{(\pi_1^{(5)})} = 0.5$. If the internal nonconformity scores

are correlated across dimensions the benefits of **Max-Rank** are likely to outweigh the cost of choosing the next larger order statistic as the threshold in the score space. In addition, the difference disappears for $n \rightarrow +\infty$.

Dimension Separability Assumption 1 clarifies the scope of the corrected construction. The assumption requires that the j th component of the internal score depends on the candidate label only through y_j . This property holds for many common score functions used in multi-target regression, including absolute residuals, normalised residuals (Papadopoulos et al., 2008), and CQR (Romano et al., 2019). The assumption is needed because the corrected finite-threshold implementation produces a Cartesian product of dimension-wise constraints. If the internal score is not dimension-separable, then changing one dimension of $\mathbf{y}$ may affect several components of the score vector. In that case, the resulting prediction region may no longer be representable as a simple hyperrectangle obtained from marginal thresholds. For genuinely multivariate scores, such as the ones discussed by Dheur et al. (2025), the corrected rank-to-score conversion would require a different geometric construction.

Experimental Limitations The experiments in this paper isolate the calibration mechanism by sampling residuals directly from a multivariate normal distribution. This design is useful because it allows us to study the finite-sample behaviour of the correction without confounding effects from model training. It also makes it possible to control the dependence between output dimensions directly.

However, this setup is still represents an idealised setting. In real multi-target regression problems, the residual distribution may be heteroscedastic, asymmetric, heavy-tailed, or dependent on the input $\mathbf{X}^{(i)}$. Moreover, the fitted model may introduce additional structure into the residuals. A natural next step is therefore to evaluate the **Corrected Max-Rank** implementation on real multi-target regression benchmarks and with different underlying predictive models.

The current experiments also focus on absolute residual scores. The theoretical result covers a broader class of dimension-separable internal score functions, but this broader scope has not yet been evaluated empirically.

7. Conclusion

In this paper, we revisited **Max-Rank** and identified two issues in its implementation in the score space. The **Max-Rank** idea is valid in the rank space. A candidate label is included in the prediction set when its worst dimension-wise rank is not too large, and this rule in the rank space can be justified via its relation to a symmetric transductive **Max-Rank** construction. The problem appears when this valid rule in the rank space is translated into dimension-wise thresholds on the original score space. A rank threshold q does not correspond to the q th smallest internal calibration score, but rather to the $q + 1$ st, with an infinite-threshold convention when the $q + 1$ st internal calibration score does not exist. The original finite-threshold implementation therefore produces intervals that are one order statistic too small and can undercover for finite samples.

We proposed the score-space **Corrected Max-Rank** implementation, a corrected version of the **Max-Rank** implementation that replaces the dimension-wise threshold of the q th

smallest internal calibration score $\beta_j^{(\pi_j^{(q)})}$ with the $q + 1$ st smallest $\beta_j^{(\pi_j^{(q+1)})}$ when $q + 1 \leq n$, and with $+\infty$ otherwise. This correction does not require changing the significance level. Under exchangeability, dimension-separability of the internal nonconformity score function (Assumption 1), and almost surely distinct internal scores, the corrected implementation faithfully represents the valid rank-space **Max-Rank** rule and therefore satisfies the desired finite-sample coverage guarantee.

The experiments confirm the theoretical analysis. The original finite-threshold implementation can exceed the targeted error level, especially for small calibration sets and a larger number of output dimensions, while **Corrected Max-Rank** remains valid at the cost of slightly wider prediction regions.

Future Work Several directions remain for future work. First, the empirical evaluation should be extended beyond synthetic residual distributions to real multi-target regression benchmarks and different underlying predictive models, in order to assess how the **Corrected Max-Rank** implementation behaves under heteroscedasticity, heavy tails, model misspecification, and input-dependent residual structure. Second, the connection between **Max-Rank**, Copula Conformal Prediction (Messoudi et al., 2021; Zhang et al., 2023; Sun and Yu, 2024), and other dependence-aware multiple-testing corrections deserves further investigation, since similar rank-to-score conversion issues may arise in related procedures or have already been identified. Finally, future work should study adaptive or weighted variants of **Max-Rank**.

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Appendix A. Proofs

This section contains the proofs of Theorem 2 and Corollary 3 together with the lemmas supporting them.

Lemma 7 *Using the same internal nonconformity score function S_{int} and model $\hat{\mathbf{f}}$, the ranks in the **Max-Rank** procedure $R_{ICP,j}^{(i)}$ as defined in Equation (4) are bounded by the ranks in the TCP setting $R_{TCP,j}^{(i)}$ as defined in Equation (15) according to*

$$R_{TCP,j}^{(i)} \in \left\{ R_{ICP,j}^{(i)}, R_{ICP,j}^{(i)} + 1 \right\}, \quad \forall i \in \{1, \dots, n+1\}. \quad (29)$$

Proof *We need to distinguish two cases.*

- *If $B_j^{(i)} < B_j^{(n+1)}$, then $R_{TCP,j}^{(i)} = R_{ICP,j}^{(i)}$.*
- *If $B_j^{(i)} \geq B_j^{(n+1)}$, then $R_{TCP,j}^{(i)} = R_{ICP,j}^{(i)} + 1$ because the rank count includes the $n+1$ st element.*

■

Intuitively, this result stems from the observation that for any i and j , the rank $R_{ICP,j}^{(i)}$ would only change when the pair $(\mathbf{X}^{(n+1)}, \mathbf{y})$ is added to the calibration set, if $B_j^{(i)} \geq B_j^{(n+1)}$. Otherwise $R_{ICP,j}^{(i)}$ would be unaffected by the addition of the pair $(\mathbf{X}^{(n+1)}, \mathbf{y})$ to $\mathcal{D}_{cal}$.

Lemma 8 *Using the same internal nonconformity score function S_{int} and model $\hat{\mathbf{f}}$, the maximum ranks in the **Max-Rank** procedure $A_{ICP,MR}^{(i)}$ as defined in Equation (5) are bounded by the maximum ranks in the TCP setting $A_{TCP,MR}^{(i)}$ as defined in Equation (14) according to*

$$A_{TCP,MR}^{(i)} \in \left\{ A_{ICP,MR}^{(i)}, A_{ICP,MR}^{(i)} + 1 \right\}, \quad \forall i \in \{1, \dots, n+1\}. \quad (30)$$

Proof *Using Lemma 7, we derive the lower bound*

$$A_{TCP,MR}^{(i)} = \max_{j=1}^m \left(R_{TCP,j}^{(i)} \right) \geq \max_{j=1}^m \left(R_{ICP,j}^{(i)} \right) = A_{ICP,MR}^{(i)},$$

and the upper bound

$$A_{TCP,MR}^{(i)} = \max_{j=1}^m \left(R_{TCP,j}^{(i)} \right) \leq \max_{j=1}^m \left(R_{ICP,j}^{(i)} + 1 \right) = A_{ICP,MR}^{(i)} + 1.$$

■

The increase in dimension-wise rank that would be caused by the addition of the pair $(\mathbf{X}^{(n+1)}, \mathbf{y})$ to $\mathcal{D}_{cal}$ described in Lemma 7 is only propagated to the nonconformity score $A_{TCP,MR}^{(i)}$ if it occurs in the dimension j that has the largest rank value.

Lemma 9 *Using the same internal nonconformity score function S_{int} and model $\hat{\mathbf{f}}$, for any threshold $t \in \{1, \dots, n+1\}$, we define $N_{ICP,MR}(t) = |\{i \leq n : A_{ICP,MR}^{(i)} \leq t\}|$ and $N_{TCP,MR}(t) = |\{i \leq n+1 : A_{TCP,MR}^{(i)} \leq t\}|$. $N_{TCP,MR}(t)$ is related to $N_{ICP,MR}(t)$ according to*

$$N_{ICP,MR}(t-1) \leq N_{TCP,MR}(t) \leq N_{ICP,MR}(t) + 1. \quad (31)$$

Proof *From Lemma 8 it follows that*

$$N_{ICP,MR}(t-1) \leq N_{TCP,MR}(t) \quad (32)$$

because $A_{TCP,MR}^{(i)} \leq A_{ICP,MR}^{(i)} + 1$, hence if $A_{ICP,MR}^{(i)} \leq t-1 \Rightarrow A_{TCP,MR}^{(i)} \leq t$. It also follows from Lemma 8 that

$$N_{TCP,MR}(t) \leq N_{ICP,MR}(t) + 1 \quad (33)$$

because $A_{TCP,MR}^{(i)} \geq A_{ICP,MR}^{(i)}$ for $i \leq n$, hence if $A_{TCP,MR}^{(i)} \leq t \Rightarrow A_{ICP,MR}^{(i)} \leq t$ and possibly also $A_{TCP,MR}^{(n+1)} \leq t$. $\blacksquare$

Lemma 9 lets us relate the number $N_{ICP,MR}$ of nonconformity scores $A_{ICP,MR}^{(i)}$ from **Max-Rank** and the number $N_{TCP,MR}$ of nonconformity scores $A_{TCP,MR}^{(i)}$ of the transductive adaptation that are below an arbitrary threshold t .

Let $\tilde{\pi}$ be any permutation of $\{1, \dots, n\}$ that sorts the elements $\{A_{ICP,MR}^{(i)}\}_{i=1}^n$ such that $A_{ICP,MR}^{(\tilde{\pi}(1))} \leq A_{ICP,MR}^{(\tilde{\pi}(2))} \leq \dots \leq A_{ICP,MR}^{(\tilde{\pi}(n))}$ and π be any permutation of $\{1, \dots, n+1\}$ that sorts the elements in $\{A_{TCP,MR}^{(i)}\}_{i=1}^{n+1}$ in the same way.⁵ Then, for any threshold t it holds that

$$A_{ICP,MR}^{(\tilde{\pi}(k))} \leq t \iff N_{ICP,MR}(t) \geq k, \quad (34)$$

and

$$A_{TCP,MR}^{(\pi(k))} \leq t \iff N_{TCP,MR}(t) \geq k. \quad (35)$$

In words, there are at least k elements in $\{A_{ICP,MR}^{(i)}\}_{i=1}^n$ that have a value at most t if and only if the k th smallest element $A_{ICP,MR}^{(\tilde{\pi}(k))}$ is at most t .

Lemma 10 *Using the same internal nonconformity score function S_{int} and model $\hat{\mathbf{f}}$, the sorted maximum ranks of the **Max-Rank** method and the transductive adaptation are related by*

$$A_{ICP,MR}^{(\tilde{\pi}(k-1))} \leq A_{TCP,MR}^{(\pi(k))} \leq A_{ICP,MR}^{(\tilde{\pi}(k))} + 1. \quad (36)$$

Proof

5. $\tilde{\pi}$ and π are not unique if multiple elements in the respective sets they sort have the same value. Any permutation that produces a valid ascending order is admissible.

Lower bound We argue by contradiction. Suppose that $A_{TCP,MR}^{(\pi^{(k)})} < A_{ICP,MR}^{(\tilde{\pi}^{(k-1)})}$. This is equivalent to showing that

$$A_{TCP,MR}^{(\pi^{(k)})} \leq A_{ICP,MR}^{(\tilde{\pi}^{(k-1)})} - 1. \quad (37)$$

Using Equation (35) this is equivalent to showing that

$$N_{TCP,MR}(A_{ICP,MR}^{(\tilde{\pi}^{(k-1)})} - 1) \geq k. \quad (38)$$

The second inequality of Lemma 9, tells us that this is only true if

$$N_{ICP,MR}(A_{ICP,MR}^{(\tilde{\pi}^{(k-1)})} - 1) + 1 \geq k. \quad (39)$$

However, there are strictly fewer than $k - 1$ elements in $\{A_{ICP,MR}^{(i)}\}_{i=1}^n$ that are smaller than $A_{ICP,MR}^{(\tilde{\pi}^{(k-1)})} - 1$. By definition of the order statistic, at most $k - 2$ elements are strictly less than $A_{ICP,MR}^{(\tilde{\pi}^{(k-1)})}$, and therefore $N_{ICP,MR}(A_{ICP,MR}^{(\tilde{\pi}^{(k-1)})} - 1) + 1 \leq k - 2 + 1 = k - 1$, contradicting Equation (39).

Upper bound Let $t = A_{ICP,MR}^{(\tilde{\pi}^{(k)})} + 1$. Substituting in the first inequality of Lemma 9 results in

$$N_{ICP,MR}(A_{ICP,MR}^{(\tilde{\pi}^{(k)})}) \leq N_{TCP,MR}(A_{ICP,MR}^{(\tilde{\pi}^{(k)})} + 1). \quad (40)$$

From Equation (34) we know that $k \leq N_{ICP,MR}(A_{ICP,MR}^{(\tilde{\pi}^{(k)})})$, and therefore, by substituting in Equation (40)

$$k \leq N_{TCP,MR}(A_{ICP,MR}^{(\tilde{\pi}^{(k)})} + 1). \quad (41)$$

By applying Equation (35) to Equation (41) we get the Lemma's second inequality

$$A_{TCP,MR}^{(\pi^{(k)})} \leq A_{ICP,MR}^{(\tilde{\pi}^{(k)})} + 1. \quad (42)$$

■

Lemma 11 Using the same internal nonconformity score function S_{int} and model $\hat{\mathbf{f}}$, the maximum rank of the pair $(\mathbf{X}^{(n+1)}, \mathbf{y})$ for the **Max-Rank** method and its transductive adaptation are related by

$$S_{ICP,MR}(\mathbf{X}^{(n+1)}, \mathbf{y}) + 1 = S_{TCP,MR}(\mathbf{X}^{(n+1)}, \mathbf{y}). \quad (43)$$

Proof Lemma 7 states that $R_{TCP,j}^{(i)} \in \{R_{ICP,j}^{(i)}, R_{ICP,j}^{(i)} + 1\}$, $\forall i \in \{1, \dots, n+1\}$. In the case when $i = n+1$

$$R_{ICP,j}^{(n+1)} + 1 = R_{TCP,j}^{(n+1)} \quad \forall j \in \{1, \dots, m\}. \quad (44)$$

because the TCP ranks as defined in Equation (15) count the element $n+1$ which is not counted by the ICP variant defined in Equation (4). The result is obtained by applying the max over all j on both sides. ■

Proof of Theorem 2 We need to show that

$$\left\{ \mathbf{y} \in \mathcal{Y} : S_{\text{ICP}, \text{MR}}(\mathbf{X}^{(n+1)}, \mathbf{y}) \leq q_{\text{ICP}}^{(1-\epsilon)} \right\} \supseteq \left\{ \mathbf{y} \in \mathcal{Y} : S_{\text{TCP}, \text{MR}}(\mathbf{X}^{(n+1)}, \mathbf{y}) \leq q_{\text{TCP}}^{(1-\epsilon)} \right\}. \quad (45)$$

Using the result from Lemma 11 this is equivalent to

$$\begin{aligned} \left\{ \mathbf{y} \in \mathcal{Y} : S_{\text{ICP}, \text{MR}}(\mathbf{X}^{(n+1)}, \mathbf{y}) \leq q_{\text{ICP}}^{(1-\epsilon)} \right\} \\ \supseteq \left\{ \mathbf{y} \in \mathcal{Y} : S_{\text{ICP}, \text{MR}}(\mathbf{X}^{(n+1)}, \mathbf{y}) + 1 \leq q_{\text{TCP}}^{(1-\epsilon)} \right\}. \end{aligned} \quad (46)$$

Equation (46) holds if

$$q_{\text{ICP}}^{(1-\epsilon)} \geq q_{\text{TCP}}^{(1-\epsilon)} - 1. \quad (47)$$

Let $k = \lceil (1-\epsilon)(n+1) \rceil$. Using Lemma 10 we know that $q_{\text{TCP}}^{(1-\epsilon)} = A_{\text{TCP}, \text{MR}}^{(\pi^{(k)})} \leq A_{\text{ICP}, \text{MR}}^{(\tilde{\pi}^{(k)})} + 1$. Therefore, substituting $q_{\text{TCP}}^{(1-\epsilon)}$ in inequality (47) with $A_{\text{ICP}, \text{MR}}^{(\tilde{\pi}^{(k)})} + 1$, it is sufficient to show that

$$q_{\text{ICP}}^{(1-\epsilon)} \geq A_{\text{ICP}, \text{MR}}^{(\tilde{\pi}^{(k)})}. \quad (48)$$

Inequality (48) holds if

$$\lceil (1-\epsilon)(n+1) \rceil \geq \lceil (1-\epsilon)(n+1) \rceil, \quad (49)$$

which holds trivially. ■

Proof of Corollary 3 Since $S_{\text{TCP}, \text{MR}}$ treats the elements of $\mathcal{D}_{\text{cal}} \cup \{(\mathbf{X}^{(n+1)}, \mathbf{Y}^{(n+1)})\}$ symmetrically and these elements are exchangeable, the prediction region $\Gamma_{\text{TCP}, \text{MR}}^\epsilon$ is valid, i.e.

$$\mathbb{P} \left(\mathbf{Y}^{(n+1)} \in \Gamma_{\text{TCP}, \text{MR}}^\epsilon(\mathbf{X}^{(n+1)}) \right) \geq 1 - \epsilon. \quad (50)$$

From Theorem 2 we have $\Gamma_{\text{ICP}, \text{MR}}^\epsilon(\mathbf{X}^{(n+1)}) \supseteq \Gamma_{\text{TCP}, \text{MR}}^\epsilon(\mathbf{X}^{(n+1)})$, and therefore

$$\begin{aligned} \mathbb{P} \left(\mathbf{Y}^{(n+1)} \in \Gamma_{\text{ICP}, \text{MR}}^\epsilon(\mathbf{X}^{(n+1)}) \right) &\geq \mathbb{P} \left(\mathbf{Y}^{(n+1)} \in \Gamma_{\text{TCP}, \text{MR}}^\epsilon(\mathbf{X}^{(n+1)}) \right) \\ &\geq 1 - \epsilon, \end{aligned} \quad (51)$$

since the TCP variant of the rank-space Max-Rank rule is valid. ■